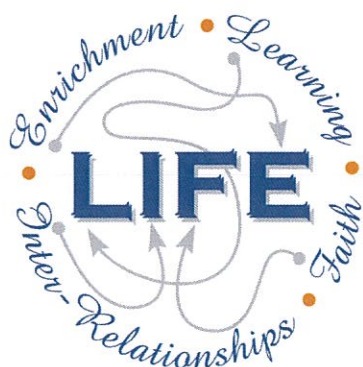




Holy Cross College

Semester 1 Examination 2012

Question/Answer Booklet

**Year 10 Science
(Modified)**

Please place your student identification label in this box

Student Name

Student's Teacher

SOLUTIONS

Time allowed for this paper

Reading time before commencing work: 10 minutes

Working time for paper: 90 minutes

Materials required/recommended for this paper

To be provided by the supervisor

This Question/Answer Booklet
Data Sheet

To be provided by the candidate

Standard items: pens, pencils, eraser, correction fluid, ruler, highlighters.

Special items: non-programmable calculators satisfying the conditions set by the School Curriculum and Standards Authority for this course.

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Suggested working time (minutes)	Marks available	Percentage of exam
Section One: Multiple Choice	40	40	30	40	31
Section Two: Short Answer	13	13	45	78	61
Section Three: Comprehension	1	1	15	10	8
				128	100

Instructions to candidates

- The rules for the conduct of examinations at Holy Cross College are detailed in the College Examination Policy. Sitting this examination implies that you agree to abide by these rules.
- Write your answers in this Question/Answer Booklet.
- Working or reasoning should be clearly shown when calculating or estimating answers.
- You must be careful to confine your responses to the specific questions asked and to follow any instructions that are specific to a particular question.
- Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.
 - Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
 - Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.

Achievement Standards

The transmission of heritable characteristics from one generation to the next involves DNA and genes.

The atomic structure and properties of elements are used to organise them in the Periodic Table.

SECTION 1 MULTIPLE CHOICE (40 marks)

Choose the best answer for each question in the Multiple Choice Question booklet.

Mark your selection with a cross.

- | | | | | | | | | | |
|----|----------------|----------------|----------------|----------------|----|----------------|----------------|----------------|----------------|
| 1 | (a) | (b) | (c) | (d) | 26 | (a) | (b) | (c) | (d) |
| 2 | (a) | (b) | (c) | (d) | 27 | (a) | (b) | (c) | (d) |
| 3 | (a) | (b) | (c) | (d) | 28 | (a) | (b) | (c) | (d) |
| 4 | (a) | (b) | (c) | (d) | 29 | (a) | (b) | (c) | (d) |
| 5 | (a) | (b) | (c) | (d) | 30 | (a) | (b) | (c) | (d) |
| 6 | (a) | (b) | (c) | (d) | 31 | (a) | (b) | (c) | (d) |
| 7 | (a) | (b) | (c) | (d) | 32 | (a) | (b) | (c) | (d) |
| 8 | (a) | (b) | (c) | (d) | 33 | (a) | (b) | (c) | (d) |
| 9 | (a) | (b) | (c) | (d) | 34 | (a) | (b) | (c) | (d) |
| 10 | (a) | (b) | (c) | (d) | 35 | (a) | (b) | (c) | (d) |
| 11 | (a) | (b) | (c) | (d) | 36 | (a) | (b) | (c) | (d) |
| 12 | (a) | (b) | (c) | (d) | 37 | (a) | (b) | (c) | (d) |
| 13 | (a) | (b) | (c) | (d) | 38 | (a) | (b) | (c) | (d) |
| 14 | (a) | (b) | (c) | (d) | 39 | (a) | (b) | (c) | (d) |
| 15 | (a) | (b) | (c) | (d) | 40 | (a) | (b) | (c) | (d) |
| 16 | (a) | (b) | (c) | (d) | | | | | |
| 17 | (a) | (b) | (c) | (d) | | | | | |
| 18 | (a) | (b) | (c) | (d) | | | | | |
| 19 | (a) | (b) | (c) | (d) | | | | | |
| 20 | (a) | (b) | (c) | (d) | | | | | |
| 21 | (a) | (b) | (c) | (d) | | | | | |
| 22 | (a) | (b) | (c) | (d) | | | | | |
| 23 | (a) | (b) | (c) | (d) | | | | | |
| 24 | (a) | (b) | (c) | (d) | | | | | |
| 25 | (a) | (b) | (c) | (d) | | | | | |

SECTION 2

SHORT ANSWER (78 marks)

Attempt all questions. Write your answers in the spaces provided.

1. In the table below, choose the correct statement for each word given. *Write the letter of the correct statement next to the word.*

WORD	CORRECT LETTER	STATEMENT
dominant	B	A - how an organism looks due to its genes.
genotype	F	B - hides the other gene in a pair (e.g. T).
gamete	G	C - see a new characteristic in an organism.
phenotype	A	D - same genes in a pair (e.g. TT, tt).
pure-bred	D	E - gene that is hidden by another gene (e.g. t).
incomplete dominance	C	F - the genes for a particular characteristic.
hybrid	H	G - sex cell of an organism.
recessive	E	H - different genes in a pair (e.g. Tt)

(8 marks)

2. Complete the table below for the parts of the cell as shown in the diagram below. Choose the correct number for the name of the part.

Choose from the following names - *not all parts are shown in the diagram.*

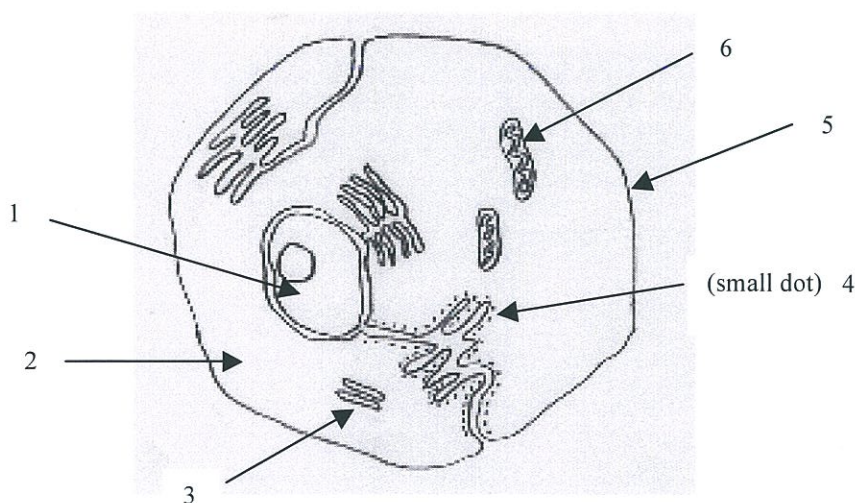
Nucleus
Ribosome
Vesicles

Cell Membrane
Mitochondria
Centrioles

Cell Wall
Endoplasmic Reticulum
Cytoplasm

Nucleolus
Golgi Complex

Number	Name
1	Nucleus
2	Cytoplasm
3	Centriole
4	Ribosome
5	Cell Membrane
6	Mitochondrion



(6 marks)

3. In the table below, choose the correct statement for each part of a cell. *Write the letter of the correct statement next to the word.*

CELL PART	CORRECT LETTER	STATEMENT
centriole	C	A - produces protein for the cell.
mitochondria	D	B - controls all cell functions and contains DNA.
cell membrane	F	C - produces spindle fibres.
nucleus	B	D - produces energy for the cell
ribosome	A	E - provides strength for plant cells.
cell wall	E	F - allows food and oxygen into a cell and wastes out.

(6 marks)

4. Carefully examine the photo of a young animal (centre) showing the genetic condition called albinism.



- (a) Describe the **phenotype** of an animal with albinism.

white fur

(1 mark)

- (b) The young lion cub inherits all its features from its parents. As you can see in the picture, the lion cub shows the condition but neither of its parents (the male and female lions in the picture) is albino.

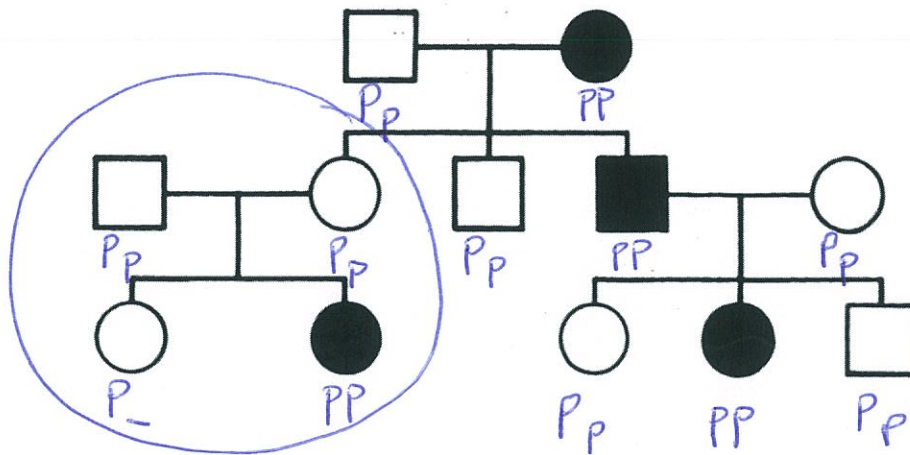
Suggest **one reason** that explains why the phenotype for albinism seen in the lion cub is not visible in either of the lions parents.

- Mutation occurred in one of the parents.
- Albino coat is a recessive characteristic.

(Either OK).

(1 mark)

5. The pedigree below shows the occurrence of polydactyly (more fingers or toes than normal) in three generations of a family.



- (a) How many people in total show the disorder in this family? 4 (1 mark)
- (b) How many people with the disorder are female? 3 (1 mark)
- (c) Is this a dominant or a recessive trait? Explain your answer.
(Circle the part of the pedigree you are considering in your answer.)

Recessive. (1)

A set of parents don't have the disorder but one child does. (1)

(2 marks)

- (b) Determine the genotypes of the individuals in the pedigree. (Remember to work from the bottom up.)

Write the genotypes neatly and clearly on the diagram.

If the second allele can't be determined, use a blank space.

e.g. A₋ or T₋ (depending on the letter you choose).

(3 mistakes = 1 mark off).

(4 marks)

5. In guinea pigs, rough coat (R) is dominant to smooth coat (r). Two hybrid rough coated guinea pigs (Rr) are mated. Determine the **genotype** and **phenotype** of the possible offspring. Include percentages to show the probabilities. Show all of your working.

(1)

♀ \ ♂	R	r
R	RR	Rr
r	Rr	rr

(1)

Parents: Rr x Rr
♂ ♀

Genotype: RR Rr Rr rr (1)

Phenotype: rough smooth (1)

Percentages: 75% 25% (1)

(5 marks)

6. In snapdragon flowers, red colour (RR) is **incompletely dominant** to white colour (WW). Hybrid plants produce pink flowers (RW).

- (a) Two hybrid **pink flowered** plants are crossed. Show the possible genotype and phenotype of their offspring.

(1)

♀ \ ♂	R	W
R	RR	RW
W	RW	WW

(1)

Parents: RW x RW
♂ ♀

Genotype: RR RW RW WW (1)

Phenotype: red pink white (1)

(4 marks)

- (b) In some plants, flowers show **co-dominance**. For example, crossing a red flower with a white flower produces red flowers with white parts, or vice versa.

Explain how co-dominance is different to incomplete dominance.

Co-dominance - see both characteristics. (1)

Incomplete dominance - see a new characteristic. (1)

(2 marks)

7. (a) Next to each phase of *mitosis*, write down the ***name of the phase*** and a ***brief description*** to explain what is occurring.

You can choose from the following, though not all stages are mentioned.

metaphase

telophase

interphase

anaphase

prophase

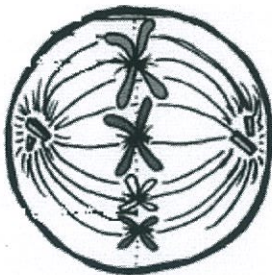


Name:

Prophase (1)

Description:

Chromosomes become visible and copy themselves. (1)

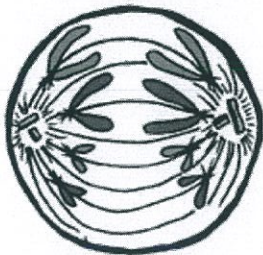


Name:

Metaphase (1)

Description:

Chromosomes line up across the middle of the cell. (1)



Name:

Anaphase (1)

Description:

Chromosomes are pulled apart by spindle fibres. (1)



Name:

Telophase (1)

Description:

Cell pinches off into 2 new cells. (1)

(8 marks)

- (b) What is the purpose of mitosis in the human body?

Growth + repair (1)

(1 mark)

- (c) Which cells of the body undergo mitosis?

All cells except reproductive cells. (1)

(1 mark)

8. (a) Who determines the sex of any children born into a family - the father (XY) or mother (XX)? Give a reason for your answer.

Father. (1)

He has the Y chromosome that determines someone being a boy. (1)

(2 marks)

- (b) A family has three girls born over five years and the mother is pregnant with a fourth child. What is the chance of the unborn child being a female?

Justify your answer by setting out a genetics calculation using a Punnett square.

Parents: $XY \times XX$

$\begin{matrix} \text{♀} \\ \text{♂} \end{matrix}$	X	Y
X	XX	XY
X	XX	XY

Genotype: $\underbrace{XX \quad XX}_{\text{female}} \quad \underbrace{XY \quad XY}_{\text{male}} \quad (1)$

Phenotype: $\text{female} \quad \text{male} \quad (1)$

Percentages: $50\% \quad 50\% \quad (1)$

(4 marks)

$\therefore 50\%$ of having a girl.

9. In the table below, choose the correct meaning for each word given. *Write the letter of the correct meaning next to the word.*

WORD	CORRECT LETTER	MEANINGS
neutron	F	A - a row of the periodic table.
period	A	B - building block of matter.
molecule	D	C - two or more elements chemically combined.
compound	C	D - two or more atoms chemically combined.
group	E	E - column of the periodic table.
atom	B	F - particle with no charge.

6
(3 marks)

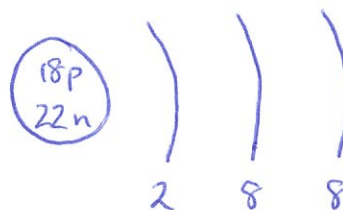
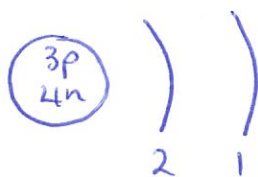
10. Complete the following table, using the periodic table to assist.

Symbol	Number of Protons	Number of Neutrons	Number of Electrons
$^{19}_{9}\text{F}$	9	10	9
$^{39}_{19}\text{K}$	19	20	19
$^{24}_{12}\text{Mg}$	12	12	12

(1 mark each column).

(3 marks)

11. Draw the electron configuration diagrams for the following elements. **Remember to include the particles in the nucleus.**



Nucleus - 1 mark

Shells - 1 mark

(4 marks)

12. Classify the following elements as **alkali metals**, **transition metals**, **metalloid (semi-metals)**, or **halogens** or **noble (inert) gases**.

Xe Te Cs Mn F Li I Ne Na Ni Ge Hg

Halogen	Noble (Inert) Gas	Alkali metal	Transition Metal	Metalloid (Semi-Metal)
F	Xe	Cs	Mn	Te
I	Ne	Li	Ni	Ge
		Na	Hg	

($\frac{1}{2}$ mark each).

(6 marks)

13. Write the symbols or names for the following elements.

ELEMENT	SYMBOL
chlorine	Cl
potassium	K
gold	Au
argon	Ar
manganese	Mn
zinc	Zn
calcium	Ca
tungsten	W
tin	Sn
boron	B

($\frac{1}{2}$ mark each).

(5 marks)

SECTION 3 COMPREHENSION (10 marks)

Read the article and then answer the questions that follow. Write your answers in the spaces provided.

Many scientists contributed to the development of the Periodic Table. The tables below summarise some information about them.

Scientist	Lived	Scientific contributions	Other facts
Antoine-Laurent de Lavoisier	1743–93	• Did not discover any new elements • Identified and named the existing 33 elements • Proved that mass was not destroyed in a reaction • Proved that rusting involved oxygen bonding to iron (increasing its mass) • Proved that combustion needed oxygen	• Nationality: French • Was very wealthy; worked as a tax collector • Had 13 000 beakers in his laboratory • Was executed by guillotine after the French Revolution • His statue in Paris was made with a 'spare' head from another statue
John Dalton	1766–1844	• Revived the Ancient Greek idea of atoms • Gave each element a pictorial symbol, which is not used today • Predicted atomic masses • Developed laws that explain how gases behave • Showed that compounds were made of elements in definite mass proportions	• Nationality: English • Came from a very poor family • Was put in charge of his small school when he was 12 years old • Still taught mathematics at a primary school when he was old and famous
Jons Jacob Berzelius	1779–1848	• Used atomic mass to arrange the elements • Gave the elements symbols that we still use today, based on their Greek or Latin names • Showed that compounds were made of elements in definite mass proportions • Was the first to classify substances as organic (containing C) or inorganic (no C)	• Nationality: Swedish • Used superscript numbers instead of subscript numbers in chemical formulas — wrote water as H²O instead of H₂O
Johann Dobereiner	1780–1849	• Placed elements into 'triads' (families of three)	• Nationality: German

Scientist	Lived	Scientific contributions	Other facts
John Newlands	1837–98	- Used atomic mass to arrange elements - Noticed every eighth element 'harmonising' (forming patterns) - Placed elements in periods of 7 elements	- Nationality: English - Called his arrangement 'the law of octaves' after musical octaves
Lothar Meyer	1830–95	- Constructed a table very similar to Mendeleev's - Finished his periodic table before Mendeleev - Published his table after Mendeleev	- Nationality: German - Is regarded as a 'father' of the periodic table
Dmitri Ivanovich Mendeleev	1834–1907	- Used atomic mass to arrange elements - Like Newlands, arranged the elements in periods of 7 - Arranged related elements in columns like Dobereiner's triads - Left gaps for undiscovered elements (e.g. Ge)	- Nationality: Russian - Refused to accept that electrons and radiation and radioactivity existed - Has a radioactive element named after him—mendelevium!
William Ramsay	1852–1916	- Discovered the noble, or inert, gases - Set them as a separate group	- Nationality: Scottish - Received the Nobel Prize in Chemistry
Henry Moseley	1887–1915	- Used atomic number, not mass, to arrange elements - Produced the periodic table still in use today	- Nationality: English - Was shot through the head and killed in World War I at Gallipoli

harmonise (v) to be similar in some way

octave (n) a group of eight notes in music

Use the information in the tables to answer the following questions.

1. Which scientist were:

- Scottish? Ramsay (1)
- Swedish? Berzelius (1)
- French? Lavoisier (1)

(3 marks)

2. Who received a Nobel Prize and for which branch of science?

Ramsay - Chemistry
(1) (1)

(2 marks)

3. Name the scientists that did the following:

- Developed the gas laws. Dalton (1)
- Left gaps in his table. Mendeleev (1)
- Developed the idea of "triads". Dobereiner (1)
- Proved that iron got heavier as it rusted. Lavoisier (1)
- Showed that compounds were made of elements in ~~fixed~~ proportions. Dalton / Berzelius
definite mass (1)

(5 marks)

END OF EXAMINATION

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